METROPOLITAN CHALLENGES - YMS30812

GOLDEN AGE TO GREEN AGE

THE LACK OF **TREE DIVERSITY** AND ITS IMPLICATIONS FOR **ECOSYSTEM SERVICES** AND **ECOLOGICAL RESILIENCE** IN THE HISTORIC CENTER OF AMSTERDAM

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07/11/2025
METROPOLITAN ANALYSIS DESIGN AND ENGINEERING

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Abstract

Urban trees play a critical role in supporting resilient and livable cities by providing a wide range of ecosystem services, including air filtration, urban cooling, carbon sequestration, and biodiversity support. Street trees in the city center of Amsterdam show a notably low diversity, with a dominance of the genus *Ulmus* (elm). Considering exacerbating climate change, this low diversity raises concerns about resilience and service provisioning. This study explores the state of tree diversity, its spatial and historical contexts, and how it relates to ecosystem services and ecological resilience. A twofold methodological approach combines spatial and regression analyses. The spatial analysis demonstrates the often-overlooked competition for underground space between trees and energy infrastructure, which complicates efforts to expand and diversify the urban forest. The regression analyses provide insights on the relationship between ecosystem services and ecological resilience on the one hand, and biodiversity on the other hand. The findings suggest that higher levels of biodiversity correlate with enhanced ecosystem service provisioning, as well as resilience of trees. Multifunctional urban ecosystems are essential, in which diversity plays a key role in supporting the robustness of the urban forest, and consequently in providing critical services to people. This research offers valuable insights for cities and urban planners, helping them effectively manage, maintain, and reap the benefits of the urban forests of the future.

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1. Introduction

Trees in Amsterdam are not growing well, which poses increasing environmental concern (Gemeente Amsterdam, 2024c). This decrease in growth has been attributed to environmental factors such as drought and heat, which are expected to exacerbate in the future due to climate change (Stott, 2016). Trees have long been a defining element of the city's identity. Historically, their primary function was aesthetic, aiming at creating pleasant parks and a uniform, orderly appearance (Gemeente Amsterdam, 2020). Over the last decades, a wide range of other services provided by street trees have increasingly become evident, such as urban cooling, carbon sequestration, water retention, noise reduction, biodiversity support, human well-being improvement (Permavoid. 2020). Street trees are now regarded as essential components of a healthy, resilient, and livable city.

Amsterdam is one of the greenest cities in the world, hosting an abundance of trees. However, one area of the city that notably lacks greenery is the center – where scarce greenspaces lack diversity. Diversity of species is essential for the resilience and thus the long-term flourishing of an ecosystem (Roebuck et al., 2022). Comparing tree diversity in Amsterdam to the seminal diversity guideline by Santamour (1990), which sets thresholds for genus and species diversity, reveals a notable area of concern. While Amsterdam as a whole meets these thresholds – unlike many other cities – the city center scores remarkably low, falling short for both genus and species diversity. A large share of the tree population in Amsterdam consists of the genus *Ulmus* (elm) (Galle et al., 2021). Such dominance of one species can result in a reduction in the ecosystem's adaptive capabilities and resistance to diseases (Roebuck et al., 2022). The lack of diversity could potentially pose a threat not only to the urban forest itself, but in turn also to the critical services that the trees provide to the people in Amsterdam. Therefore, it is critical that the effect of tree diversity – or the lack thereof – is studied.

Although ecosystem services appear to be highly dependent on diversity, the relationship is not always straightforward. From the perspective of a specific service, having a diverse range of species is not necessarily beneficial; instead, what matters is a species' ability to provide that service (Ridder, 2007). Balancing the resilience gained from diversity with the service gained from species is an important aspect of effective management of urban green space.

While extensive research has been done on the benefits of green spaces within an urban area, relatively little attention has been paid to how species dominance influences the range and quality of ecosystem services provided by trees. Understanding this relationship is crucial in effective management of the urban forest and balancing resilience and services. Therefore, this paper aims to examine the tree diversity in the city center of Amsterdam and its relationship to ecological resilience and the ecosystem services they provide. The following research question is utilized:

‘*To what extent is street tree diversity associated with ecological resilience and ecosystem service provisioning in the context of the spatially constrained city center of Amsterdam?*’

Three sub-questions are addressed, which contribute to answering the main research question:

1. ‘*What is the state and context of tree diversity in the center of Amsterdam?*’

2. ‘*To what extent does the ecological resilience of urban trees correlate to the tree species diversity in the city center of Amsterdam?*

3. ‘*To what extent are the ecosystem services provided by urban trees correlating to the tree species diversity in the city center of Amsterdam?*’

The conceptual and theoretical framework presents an extensive literature review and conceptualization of ecosystem services, ecological resilience, biodiversity, and their connection, to effectively answer the questions posed.

Amsterdam's city center was selected as the case study for this research. Amsterdam has a broad range of tree species; however, Amsterdam’s city center significantly lacks tree diversity in terms of both genes and species. The dense and historic character of this area presents specific challenges, making it a particularly insightful context for study. In contrast to more 'forgiving' contexts, the management of the city center’s ecosystem requires a high degree of precision and careful attention to detail.

In the face of climate change and the resulting extreme weather events putting pressure on urban ecosystems, understanding the relationship between street tree diversity and both the service they provide and their resilience is essential. This research aims to offer valuable insights for cities and urban planners, helping them to effectively manage, maintain, and reap the benefits of the urban forests of the future.

2. Conceptual and Theoretical Framework

2.1 Ecosystem Services

Cities depend on ecosystems outside of their boundaries but can also benefit significantly from ecosystems within their boundaries. The term ‘ecosystem services’ refers to any benefit humans can obtain from a natural ecosystem (Costanza et al., 1997). These services are often divided into three primary functions: provisioning, regulating, and cultural. Provisioning services are the physical, usable products that the ecosystem provides. Regulating services are about how trees maintain balance in the urban environment. Cultural services are about how trees enrich our experience of a city by making it more beautiful, meaningful, and emotionally restorative. Within the context of urban ecosystems and specifically street trees, the services provided generally do not include provisioning functions such as food or timber. Street trees primarily concern regulating and cultural functions (Salmond et al., 2016), which will be discussed in the following section.

Cities are significantly hotter than their surrounding hinterlands, especially during the night. This phenomenon is known as the urban heat island effect and it makes cities particularly vulnerable to heat, notably in the light of climate change (Oke, 1984; Heaviside et al., 2017). During hot periods, vegetation has a climate regulating effect. By providing shade and cooling air through evo-transpiration it can significantly lower temperatures of an area, thus making them a valuable climate regulating asset to a city (Gill et al., 2007; Picot, 2004; Silva et al., 2010). On a local scale, street trees provide shade to the street they are situated in, and to buildings within their proximity, thereby lowering indoor temperatures as well (Mavrogianni et al., 2014). In the context of the water-rich center of Amsterdam, street trees might have extra cooling potential due to the cooling synergy between vegetation and water (Zhou et al., 2022).

Street trees can regulate and improve air quality by absorbing pollutants and enhancing their deposition. The capacity of capturing pollutants from the air is dependent on the tree species and its leaf surface (Nowak et al., 2001). However, there is little evidence for the direct effect of this regulation function on physical health (Salmond et al., 2016). The relationship between street trees and air quality is not positive per se. A study by Fry et al. (2025) found that pollutant concentrations were higher in streets with increased canopy cover. It is hypothesized that trees act as a sort of dome that 'traps' pollutants in the street and prohibits them from dispersing into the atmosphere.

Vegetation has been shown to have the capability to reduce urban noise (Fang & Ling, 2004). Noise can cause certain distresses to humans such as interference of tasks, disturbance of social behavior and annoyance (Mucci et al., 2000). Furthermore, the presence of natural sounds conceals the perception of anthropogenic sounds (De Coensel et al., 2011; Dzhambov & Dimitrova, 2014). In other words, by adding natural sounds, such as rustling leaves or singing birds, people are less affected by anthropogenic noise, such as traffic noise, which increases the overall pleasantness of the soundscape.

Urban trees offer a water regulatory service during rain events by reducing and delaying water runoff (Livesley et al., 2016), which can be particularly important in urban areas because of the extensive presence of impervious surfaces. Tree roots help absorb rainwater. This process functions optimally during light to moderate rain events. Additionally, tree leaves intercept raindrops, preventing them from hitting the ground directly. This delays and spreads out the runoff, while simultaneously allowing for some of the water to evaporate before it ever reaches the soil (Zabret & Šraj, 2019). Street trees cannot stop entire floods by themselves, but they do reduce and delay rainwater runoff.

Biodiversity can be described as an ecosystem service in conservation perspectives (Mace, Norris, & Fitter, 2012). Here, an important distinction is made from tree diversity, which concerns the diversity of trees in an area specifically. Biodiversity as a service refers to the extent to which a tree species promotes biodiversity by providing habitat, food resources, or other favorable conditions for other organisms (Mace, Norris, & Fitter, 2012). For instance, certain native species may provide habitat for a wide range of insects, birds, and fungi, thereby enhancing local ecosystem complexity and diversity.

Trees hold different meanings for different people. The cultural value provided by street trees is less tangible than their regulating services. Cultural services often are associated with the stress-reducing potential of greenery as well as their aesthetic qualities (Salmond et al., 2016). The experiencing nature is widely studied and conclusively good for well-being (Russell et al., 2013). De Vries et al. (2013) found that the quality and quantity of green space is positively related to both mental and physical health outcomes. The cultural effect of street trees specifically remains understudied. Case studies have found generally positive attitudes towards street trees (Schroeder et al., 2006), and positive effects of amount of street trees on perceived attractiveness of a walking route (Borst et al., 2008).

Ecosystem services provided by street trees are essential to the functioning of cities. The flourishing and health of the urban forest is therefore of great importance. One of the key factors for maintaining this health is biodiversity (Harrison et al., 2014).

2.2 Urban Biodiversity

Biodiversity refers to the variability and richness of life and nature on earth and can be examined at a genetic, species and ecosystem level. Genetic diversity describes the variation in genes within the same species, which enhances a species' resilience to environmental pressure like diseases and climate change. Secondly, species diversity considers the number, richness and distribution of species within an ecosystem. Generally, this is the level of focus in research. Lastly, ecosystem diversity refers to the variety of ecosystems within a region, emphasizing how different landscapes each uniquely contribute to the world's biodiversity (Thomas, 2025; Center for Sustainable Systems;

University of Michigan, 2025). Together, these three levels illustrate the complex nature of biodiversity and how it serves as a fundamental concept for understanding ecological interactions and its significance for maintaining ecosystems.

A rich biodiversity is crucial for maintaining ecosystems on earth because the interactions between all species provide essential resources like water, food, medicines and shelter (What is biodiversity, sd). *Urban biodiversity*, in turn, describes the variety of life and nature within an urban context. Emphasizing this distinction is significant as cities provide a very different foundation for species development compared to natural landscapes and biodiversity is generally lower in urban areas (Beninde et al., 2015; Guo et al., 2019). Patterns of urban biodiversity are largely influenced by both physical environmental and socio-economic factors. Environmental factors, including soil conditions, temperature variation and water availability, determine whether and which species can thrive. Socio-economic factors such as extensive land use, dense building blocks and abundance of paved surfaces affect the distribution and composition of tree structures (Kendal et al., 2014). In heavily human managed and cultivated cities, these factors often cause fragmenting ecosystems and low green space density, presenting challenges for maintaining biodiversity in the urban context (Biodiversity loss poses direct threat to economy, 2024; Thomas, 2025).

There are often new species introduced in these environments, including invasive ones. These species not only outcompete other sensitive species in the city context, but similar patterns occur globally, which results in biotic homogenization. Urbanization has been associated with a loss of roughly one-third of native species that once occurred that region. Moreover, species richness tends to decline moving from the rural outskirts towards the urban core. While the number of non-native species increases closer to the city center (Elmqvist et al., 2016).

The most common tree species in Amsterdam is the Elm, which therefore represents a significant share of the city's tree network (Gemeente Amsterdam, n.d.). Lack of tree species diversity can be a problem for the resilience of the urban green structure. A specific tree species has a certain set of characteristics which makes them adaptable in specific conditions such as heat, drought, or flooding. If an urban forest consists of mainly the same tree species, the whole area is vulnerable for the same circumstances, making them less resistant (Paulin, M., 2019; Roebuck et al., 2022).

To manage and monitor tree biodiversity in the urban area, various benchmarks have been proposed throughout the years, based on the relative abundance of the most common species. Some of these measures were quite strict, but this introduced other risks. To comply to these guidelines, stronger and more abundant species were replaced by weaker trees that were underperforming in the urban forest, reducing the overall biodiversity after all. A more realistic benchmark was introduced by Santamour (1990), proposing the 10/20/30 'rule of thumb', which up until today has been a widely used metric. These guidelines state that within an urban forest, no more than 10% of trees should belong to any species, no more than 20% to any genus and no more than 30% to any family.

Generally, biodiversity is measured by the species variability of plants, animals and microorganisms in a certain area. A common statistical tool that is often used for measuring biodiversity in a certain community is the *Shannon index*. This tool uses both the number of species and the evenness of their distribution. Interestingly, the relative abundance of the most common species has been shown to be a great predictor for biodiversity measured by the Shannon index (Kendal, Dobbs, & Lohr, 2014). The outcome helps preserve certain minorities in ecosystems or identify high biodiversity areas, as well as see changes over the years in certain areas (Kharoliwal & Shrivastava, 2025).

2.3 Urban Ecological Resilience

Resilience of urban trees is important to examine, as cities increasingly face environmental challenges, exacerbated by climate change (Huff et al., 2020).

Holling (1973) defined ecological resilience as the ecosystem's capacity to withstand disturbances without fundamental alteration of its core processes and structures or completely collapsing. This departed from the scientific literature of that time, which primarily focused ecosystem stability, referring to a single equilibrium and its ability to return to that stable state after disruption. More recent research has further refined these insights, depending on assumptions about whether systems are characterized by the presence of one or more states of equilibrium. Holling (1973) emphasized that ecosystems with multiple stable states can overcome fluctuation without necessarily failing and thus the magnitude of disturbances a system can endure. Other authors have defined resilience as the time it takes to get back to the previous state after a disruption of the ecosystem (Huff et al., 2020). More recent theories focus on adaptive capacity, emphasizing the resilience is shaped by mechanisms that allow an ecosystem to adjust to environmental changes (McPhearson et al., 2019). Equilibrium theories are often inadequate for capturing the complexities of the urban environment, where it is so much changed and influenced by natural disturbances as well as. In such rapidly changing contexts, no single equilibrium would exist, highlighting the relevance of non-equilibrium ecology paradigm. This approach is necessary for a comprehensive understanding of the urban environment and its influence on ecological resilience (Mori, 2011). Other research also indicates that for overall urban resilience to thrive, a more social-ecological perspective should be considered, that incorporates both human and ecological urban systems (Huff et al., 2020). Nevertheless, this paper focuses on just the ecological dimension of urban resilience, as it is most relevant for addressing the main research question.

These foundational theories are often based on rural environments and not always specifically applied to the urban context. Nevertheless, ecological resilience is at least critically equal in cities, as it plays a vital role in protecting human populations from environmental risks like heats, flooding and natural hazards (Feng et al., 2024).

However, this urban resilience is largely threatened by human interventions, particularly through landscape fragmentation resulting from rapid urbanization and industrialization, as well as other disturbance regimes and poor soil conditions (Scharenbroch et al., 2017; Huff et al., 2020). Therefore, the urban landscape is both the medium for monitoring resilience, as well as an indicator for its overall state (Wang et al., 2024).

Trees are already visibly suffering from impacts of drought and heat in rural environments, and these effects are amplified in the urban context, research state an average temperature increase of 3 °C on summer days in the Amsterdam, in comparison with the surrounding hinterlands. (Koomen & Diogo, 2017; Janssen, ten Napel, & van Schayck, 2023). Despite the rising demand for climate-resilient trees, limited research exists on which trees thrive under certain conditions. Many of the existing studies on this are outdated and no longer reliable, leading to urban tree managers to rely on intuition and experiencing rather than research-based knowledge, an approach that doesn't always create successful results, especially in a changing climate (Janssen, ten Napel, & van Schayck, 2023).

The well known 'rule of thumb' by Santamour (1990), serves as a guideline for a healthy distribution of tree species, aiming at developing a resilient and diverse tree network within the urban green structure. Diverse species compositions provide a broader range of adaptive capabilities, making urban forests more tolerant to different stressors and increasing the ecosystem's resistance to pests and diseases (Huff et al., 2020; Paulin, 2019; Roebuck et al., 2022).

2.4 Linking Ecosystem Services and Urban Biodiversity

Both biodiversity and ecosystem services are complex concepts, and the interplay between the two is equally so. Biodiversity can be seen as an important factor that regulates ecosystem processes, which are important in providing services. Alternatively, biodiversity can be described as a service in itself. People assign different values to biodiversity, whether in their value of specific species or in a desire for its existence, regardless of direct benefit or experience (Mace et al., 2011).

The effect of biodiversity on ecosystem services is generally thought of as evidently positive (Harrison et al., 2014). Areas with greater biodiversity typically provide a broader range of ecosystem services than less diverse environments. Biodiversity is widely understood to be significant to general ecosystem health and prosperity, and therefore essential to the provisioning of any service to humans (Jiao et al., 2021; Thomas, 2025). From this perspective, a drop in biodiversity would lead to a reduction in the services provided. However, the direct effect on ecosystem service outcomes is more ambiguous and has been the subject of academic debate. This section explores literature on the link between biodiversity and ecosystem services, explaining its complexities a how different features of species influence the quantity and quality of ecosystem services provided.

An important consideration that challenges the application of biodiversity for ecosystem service is that most services are delivered not by ecosystems, but by certain groups of species that possess specific functional traits (Ridder, 2007). Certain services are thus dependent on the presence of these species, and not the richness or abundance of species. For example, when it comes to the cooling function, the cooling potential of an ecosystem does not depend directly on species diversity, but rather on the presence of species with particularly high cooling potential. In these kinds of situations, biodiversity plays a secondary role. Furthermore, many services depend simply on the presence of trees or vegetative cover, rather than on tree species. An ecosystem's resilience is to a certain extent reliant on biodiversity. Ridder (2007) argues that once a biodiversity ‘threshold’ is met to secure resilience, biodiversity should not be of the highest priority. In this way, he reframes the precaution principle, which holds that because we cannot predict which species are critical, we must conserve all species to avoid catastrophes. If the goal of an ecosystem is strictly to provide services, it would be more sensible to manage the specific species or processes that directly supply each service, rather than protect all biodiversity. This view opposes the Santamour (1990) guidelines, which set a certain desirable threshold for general biodiversity, instead of incorporating case specific functions.

Continuing this, the (multi)functionality of species and its relationship to the quality of ecosystem services can also be explained by the 'jack of all trades effect' (Van Der Plas et al., 2016). This mechanism explores the situation where a group of various species have a moderate influence on certain ecosystem services rather than excelling in just one. Through this lens, a diverse ecosystem would serve a broad range of ecosystem services on a moderate level. This creates an overall positive relationship between biodiversity and ecosystem services, but a diverse community would never be able to increase to the highest possible level of an ecosystem service. However, this effect does emphasize the importance of diversity in providing a broad foundation of ecosystem services. Isbell et al. (2011, p. 1) add to this narrative: “although species may appear functionally redundant when one function is considered under one set of environmental conditions, many species are needed to maintain multiple functions at multiple times and places in a changing world”. Even if some species seem interchangeable for a single function in a specific environment, a variety of species is necessary to sustain multiple ecosystem functions across different times and conditions. This idea becomes increasingly relevant in light of climate change, and the ecosystem's ability to cope with extreme weather events.

Research in forest settings demonstrates that higher species richness is associated with an increase in multiple ecosystem services (Gamfeldt et al., 2013). However, both the context and the services highlighted in these studies differ substantially from the urban case discussed in this paper. Most studies on tree biodiversity in relation to their provisioning of ecosystem services in urban areas have been conducted on a city wide or regional level. However, Roebuck et al. (2022) emphasize the need for similar research on a smaller, more localized scale, as it allows for a better understanding of the site-specific biodiversity characteristics and the implementation of more targeted urban planning strategies. The nexus between biodiversity and ecosystem services is understudied within the urban context.

While biodiversity is often used to describe the variety of species in an area, this paper focuses on biodiversity as a determinant of the quality or quantity of ecosystem services in a city. Specifically, the diversity of trees in an urban area affects both the abundance and quality of ecosystem services. At the same time, various characteristics and historical aspects of the urban context shape tree biodiversity. Factors such as past land use and urban trends, current heritage conservation and infrastructure influence today's biodiversity patterns (Jiao et al., 2021).

2.5 Amsterdam City Context

Amsterdam's city center ('Centrum'), as defined by its administrative boundary, exhibits limited street tree biodiversity in both genes and species terms. Within this area, genus *Ulmus* accounts for 72% of all genera, and 47% of the tree species are *Ulmus hollandica* (Galle et al., 2021). These numbers both significantly violate the Santamour (1990) guideline, as shown in figure 1. In contrast, in the outer parts of the city, genus *Ulmus* represents only 19% of total diversity, while the most abundant species is *Platanus acerifolia*, comprising 11% of total diversity (Galle et al., 2021). These findings align with broader urban ecology literature, indicating that the city centers usually show a more homogeneous and less diverse landscape.

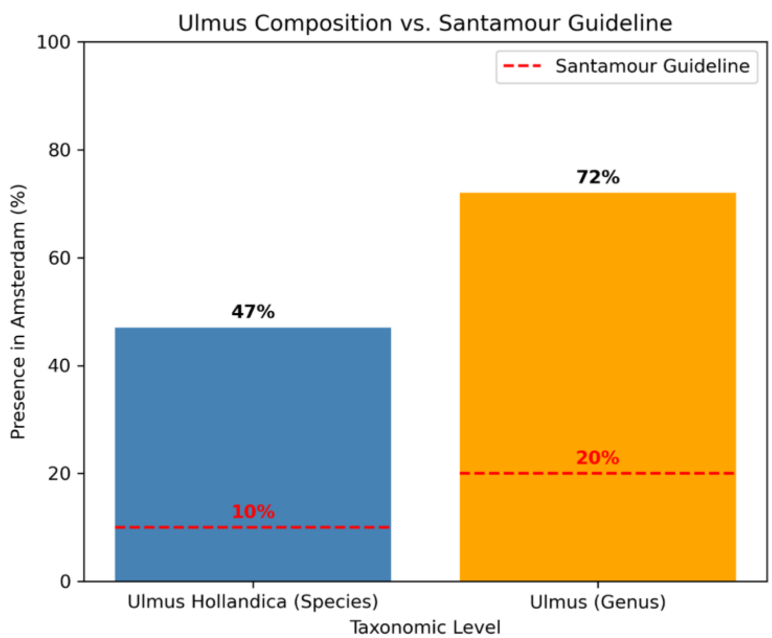

Figure 1. Ulmus genus and species composition in the center of Amsterdam

Furthermore, research shows that historic city centers such as Amsterdam are characterized by high building densities and limited green space, often restricted to the form of small parks, gardens, or residual plots. In these urban settings, the ecosystem services that are provided are restricted to recreational services (Russo et al., 2025).

The center of Amsterdam is a protected UNESCO World Heritage site and a designated green monument area. While this status preserves the city's cultural and historical environment, it limits the scope of ecological interventions necessary to strengthen biodiversity and ecosystem service provision. Combined with a high population density and limited green space availability, these factors create a complex setting where heritage protection, ecological restoration, and urban livability must coexist within tight boundaries.

Heritage protection imposes strong regulatory and institutional constraints on the implementation of green infrastructure. According to the World Heritage Canal Ring Management Plan (Gemeente Amsterdam, 2023), any physical alterations within the inner ring must maintain visual authenticity, reversibility, and material preservation of the historic fabric. These rules, while essential for protecting cultural heritage, restrict the installation of large vegetation, soil-based interventions, or façade modifications that could enhance regulating and supporting ecosystem services.

Over the centuries, Amsterdam has developed green spaces, which structure the city and give it an identity as the world-famous canals lined with elm trees. Green spaces have also played an important role in national heritage neighborhoods, and several parks are esthetically designed and planted that they have been designated as monuments. However, as Amsterdam flourishes and the population keeps growing, the city is entering a new phase of its existence and is facing major changes (Amsterdam Green Infrastructure Vision, 2025).

Population growth further intensifies the city's spatial pressures. In 2023 alone, Amsterdam's population increased by 13,553 residents (a 1.5 percent rise). Both population and population growth are not evenly distributed throughout the city; all areas around downtown are more densely populated than the rest (Open Research Amsterdam, 2023). As a result, space has become increasingly scarce across the Netherlands; nearly every square meter has a designated use, whether for living, working, nature, recreation, or another purpose. Various societal issues, such as housing construction, energy transition, nature restoration, water and soil conservation, the transition in agriculture, compete for limited space causing complex challenges, including healthy living environment, nature, water and soil (Ruimtelijke Economische Verkenning , 2024).

Among the most significant threats to Amsterdam's future is the rising sea level worsened by climate change. Approximately 55% of the Netherland's territory classified as flood prone with certain parts of the city lie beneath sea level. Although the probability of a flooding event is low, the consequences are quite high (Swinkels, 2020). Heat stress forms an additional challenge for Amsterdam's city center. During a substant heat wave in 2015, extreme heat stress was mostly found in the city center (Ellenbroek, 2023).

The municipality of Amsterdam addressed these issues by establishing Green Infrastructure Vision 2025. One of the four key principles is described as '*Biodiversity will be integrated into urban* *planning, construction, and management.'* The document emphasizes the importance of increasing urban biodiversity, recognizing that while much of the country is becoming urbanized, many plant and animal species now depend on cities to survive. Amsterdam's biodiversity has been increasing in recent years, and the municipality aims to sustain this trend by enabling plants and animals to inhabit the city as a thriving natural environment (Gemeente Amsterdam, 2020).

Under this principle, it has four practical strategies: "*First, when making new or refurbishing existing green, blue, or built-up areas, choose biodiversity enhancing designs for watersides, quay walls, road verges, parks, forests, grasslands, gardens, streets, and/or squares. Second, integrating biodiversity in the construction, renovation, and transformation of buildings will be a building regulation, Third, Ecologic maintenance of public space and watersides will become the norm by 2030 at the latest. Fourth, extending the existing ecological structure of the city and not causing any new ecological barriers as the city densifies"* (Gemeente Amsterdam, 2020).

2.6 Spatial Conflicts of Urban Trees

The Puccini Method Guideline of the municipality states that a tree requires approximately 2 meters of clear underground space to prevent damage to utility lines (Gemeente Amsterdam, 2024b). This number aligns closely with the 2.1 meter needed according to the Rekenprogramma Boommonitor for trees to live a projected lifespan of 60-year (Norminstituut Bomen, 2022). Although the clearance of spaces near trees supports a potential 60-year lifespan, the average lifespan of an urban tree in Amsterdam is estimated to be around 40 to 43 years. This 40-to-43-year lifespan average is confirmed by both the findings of Voeten (2013), cited in the research The Roots of the Problems (Brown, 2022), and the municipal advisory document on protected tree species (Gemeente Amsterdam, 2024a).

Proactive planning like the Puccini method is needed to ensure an urban tree's survival and achieve maturity in the urban landscape. Both the above ground and underground factors are crucial to consider. In the above ground, it is important to consider things like sunlight exposure, wind direction, proximity to buildings, and space needed for the canopy spread (Miron and Millward, 2024), and the below ground considerations are equally important. Yet attention to the subterranean requirements for tree growth is often overlooked in urban planning, which generally takes engineering requirements as a clear precedence over healthy tree growth (Jim, 2003; Miron and Millward, 2024).

Tree stability depends heavily on both the root system architecture and the anchorage strength of the roots (Watson, 2014). The anchorage strength is divided into four key components: the mass of the roots, the strength of the soil and depth of penetration, the resistance to failure in tension, and the length of the lever arm (Watson, 2014). As each of these components increases, the force required to uproot the tree also increases.

Despite the importance of root health, increasing the size of the root plate is extremely challenging in urban environments. Urban trees face intense and constant conflict with surrounding infrastructure, including buildings, roads, and utility installations (Morell, 1992). These conflicts frequently result in low tree performance due to the poor site conditions (Weaver & Stipes, 1998). Consequently, most trees planted in urban infrastructure have root systems that were severed during transplantation (Watson, 2014). This led to branching at the cut end and smaller regenerated roots. It may lead to a reduction of the main root diameter, with the potential effect of destabilizing trees (Watson, 2014).

Furthermore, maintaining healthy growth can be disturbed by construction activities such as soil compaction, placing fill over roots, root severing and abrupt changes in drainage pattern, which often causes tree decline and mortality (Foster, 1977; Matheny & Clark, 1998; Schoenweiss, 1982; Watson & Neely, 1995).

3. Methodology

This study uses a twofold methodological approach, combining regression and spatial analyses, in order to provide answers to the sub questions and ultimately the research question. The first approach comprises a data analysis, which was conducted through regression models. The second entails a spatial analysis and was performed to identify conflicting areas between street trees space and underground infrastructure. Furthermore, this study included an extensive literature review, covering dominant theories and concepts, as well as case studies on the topic, as presented in the conceptual and theoretical framework.

3.1 Data Collection

In assessing the biodiversity for each neighborhood in the center, data on tree diversity is collected from the *maps.amsterdam.nl* website. This site features a publicly available map of the city which includes data on every tree located within public space. For each tree, the features of interest in this study include its location, species, and genus.

The dependent variables are based on information provided by the Amsterdam municipality in the Puccini Green Handbook, a policy and technical manual for designing and managing the city's public green spaces. In the appendix of this handbook, many tree species are given a score of 1 to 4 (provided as -, +, ++, +++) based on specific resilience characteristics as well as on ecosystem service contributions. The ecosystem services include reduction of heat stress, capture of pollutants, and contribution to biodiversity. The ecological resilience characteristics include tolerance for shadow, drought, and water. Two important considerations need to be discussed concerning this data. First, the handbook provides information on three relevant ecosystem services but lacks data on several other important ones mentioned in the theoretical framework, such as water retention and noise reduction. This limits the scope of the analysis, as certain important services are not accounted for. Second, as discussed in the theoretical framework, in this context, the contribution to biodiversity refers not to the tree species itself and its inherent biodiversity value, but rather to the extent to which the species *facilitates* broader biodiversity.

Data on the control variables were collected from existing data sets, retrieved from the *Dashboard kerncijfers* website (onderzoek.amsterdam.nl/interactief/dashboard-kerncijfers) by the statistics department of the municipality of Amsterdam. Sources include 1) the *Wonen in Amsterdam* survey (WiA, 2023), conducted by the municipality. Neighborhoods are included in the published survey data only if they have at least 20 respondents. 2) The national statistics department of *Centraal Bureau voor de Statistiek* (CBS), which conducts yearly surveys on socio-economic status. 3) The *Onderzoek & Statistiek* (O&S) department of the municipality of Amsterdam.

3.1.1. Cleaning

This data was prepared and cleaned in order to ensure completeness and accuracy in regression analysis. In cases where data points were missing for neighborhoods, this was first supplemented with values from the two previously available years. Despite this, there were still missing values in the case of some independent neighborhood variables. This was addressed by calculating the average values of directly neighboring neighborhoods. For instance, if neighborhood A was missing data and adjacent to neighborhoods B, C, and D, the value is A is determined by the average of B, C, and D. While this is a slight limitation to the validity of the regression, these steps were taken to ensure a complete and inferable analysis. To prevent negative values interfering with regression, tree data from the Puccini handbook (originally -, +, ++, +++) was changed to a scale of 1-3/4. Negative values do not mean decreased ecosystem services and resilience, so this changed scale does not meaningfully impact the validity.

The data was cleaned to combine the Puccini data with the tree database. Tree species were simplified to match the two datasets, and a total of 6.997 matches were found. However, to incorporate more matches for the regression, the two biggest species missing were added to the Puccini method. The two species, (Ulmus Hollandica Vegeta, Ulmus Hollandica Commelin), where sub species are from the Ulmus genus. These subspecies were compared with the existing Ulmus species in the database to supplement the values and create a dataset with 8.980 matches.

3.1.2. Spatial Data

The data for the spatial analysis has three main components that function as bases. First, the spatial location of the trees is used from the *maps.amsterdam.nl* website. This site features a publicly available map of the city which includes data on every tree located within public space. Secondly, the 2D spatial locations of the underground sewage system from the API data Amsterdam (api.data.amsterdam.nl/v1/docs/datasets/leidingeninfrastructuur) are used for the model. Lastly, the 2D spatial location of the underground power infrastructure from Liander on PDOK (service.pdok.nl/liander/elektriciteitsnetten) is used to complete the bases. This data is confined to only the inner city of Amsterdam as a research scope.

3.2 Methodological Approach

3.2.1 Regression Analysis

Data analyses were utilized to investigate the association between street tree diversity and ecosystem services, utilizing neighborhood level (using the boundaries provided by the municipality) as the unit of analysis. Additionally, analyses on the association between tree diversity and resilience scores were run. After adjusting for missing values, the sample size at the neighborhood level is $n = 69$.

Dependent variables

For the operationalization of the dependent variables, information provided by the municipality in their Puccini Green guideline is used. This guideline elaborates data on the ecosystem services associated with specific tree species, as well as their tolerance to different environmental factors.

Different ecosystem services are graded on a scale of 1 to 3 per tree species, based on their relative performance in delivering a given service. A score of 1 represents a low contribution; a 3 indicates a high contribution. The ecosystem services covered include cooling effect, capturing pollutants, and contribution to biodiversity. The resilience characteristics include tolerance for shadow, drought, and water.

To quantify the contribution of street trees to ecosystem services, an Ecosystem Service Index (ESI) was developed for each study area. The calculation of the ESI is done as follows. The ecosystem service scores are summed up for all trees within an area and then divided by the surface area, which makes it possible to compare areas of different sizes. This approach enables the index to account for the number of trees delivering ecosystem services as well as their degree of contribution, relative to the surface of the area. The equation for the ecosystem service index (ESI) and the resilience index (RI) is expressed as:

$$ESI^{(service)} = \frac{\sum_{i=1}^{n} S_i}{A} \qquad RI^{(characteristic)} = \frac{\sum_{i=1}^{n} S_i}{A}$$

Here, *ESI* is the ecosystem service index for each separate service. *RI* is the resilience index for each separate resilience characteristic. *i* is the index for each individual tree, *n* is the total number of trees in an area, *S* is the ecosystem service score per tree (ranging from 1 to 3), and *A* is the surface area of the area in m^2.

Multiple ecosystem services were evaluated, reducing heat stress, capturing pollutants, and contributing to biodiversity. Individual indices were calculated for each service using the same formula. The same was done for the different resilience characteristics: shadow tolerance, drought tolerance, and water tolerance.

Independent variables

The operationalization of tree species diversity is determined by applying the Shannon Index (Shannon, 1948). This index is widely used in ecology as an indicator for biodiversity. It captures both the richness (number of species) and the evenness (distribution of individuals among species) within a given area. The Shannon Index was calculated on the species level for every neighborhood. The equation of the Shannon Index (H') is expressed as:

$$H' = -\sum_{i=1}^{S} p_i ln(p_i)$$

Here, p_i represents the proportion of individuals of a specific species in relation to the total number of individuals of all species. p_i is calculated by *n/N*, in which *n* is the number of individuals of a species and N is the total number of individuals. S is the species richness, meaning the total amount of distinct species. The higher the index score of an area, the more diverse it is deemed to be.

Control variables

Besides biodiversity, a range of factors could play a role in shaping ecosystem services, as well as resilience. Controlling their confounding effect is crucial for the accuracy and validity of the analysis. The included control variables are discussed in this section, along with the justification for their inclusion.

First, population density affects both demand and provision of services (Richards et al., 2019). High population density often correlates with more built-up space and thus a higher stress on vegetation (Fuller & Gaston, 2009). Dense areas are generally negatively associated with ecosystem service outcomes (Chen et al., 2019). High density may reduce the capacity for greenery to provide ecosystem services due to factors that limit their growth. These factors include a lack of rooting space due to piping, and physical damage due to dense mobility. In densely built environments, trees often experience harsher environmental factors, which can reduce their ability to tolerate and recover from weather extremes.

Second, including socio-economic status allows the model to control the potential confounding effect of income. Socio-economic status affects ecosystem services through its influence on the management of urban green space and by altering social practices and behavior, and therefore people's demand for certain services (Wilkerson et al., 2018). More affluent neighborhoods might experience better greenspace management and resources, whether intentional or not. Both the services provided, and the ecological resilience, could thus be influenced by socio-economic status, which is why it is incorporated as a control variable.

Third, pollution is included as a control variable because it can directly affect tree health, and growth which in turn influences its ecological resilience and the ecosystem services they provide. Air pollution has a negative effect on tree growth (Locosselli et al., 2019). Therefore, neighborhoods with more pollution may have lower ecosystem service scores, regardless of tree diversity. Since there is no publicly available data on pollution per neighborhood, perception of pollution is used as an indicator for pollution in the neighborhood, as they can reflect actual environmental conditions.

Fourth, the maintenance of greenspace is deemed an important factor for both ecosystem services and resilience. As with the case of air pollution, citizen perception data was used. Trees that are maintained well live longer and healthier, which could improve their capabilities to thrive during harsher weather and deliver services. Therefore, neighborhoods with well-maintained greenspaces may score higher in ecosystem services and ecological resilience independently of tree species diversity.

Fifth, in the models which utilize ecosystem services as the dependent variable, ecological resilience was included as a control variable because the ability of trees to withstand and recover from environmental stressors directly influences the level and stability of the services they provide (Isbell et al., 2011). Trees that are more resilient to weather extremes such as drought, heat, or storms are likely to maintain higher rates of ecosystem functioning. The final list of variables is depicted in Table 1.

Variable		**Source**	**Date**	**Unit**
Dependent Variables				
Ecosystem Services	Cooling Effect	Puccini Green Handbook (Gemeente Amsterdam, 2024b)	Feb 2024	Ratio cooling effect score to land area
	Reducing Pollutants	Puccini Green Handbook (Gemeente Amsterdam, 2024b)	Feb 2024	Ratio reducing pollutants score to land area
	Supporting Biodiversity	Puccini Green Handbook (Gemeente Amsterdam, 2024b)	Feb 2024	Ratio supporting biodiversity score to land area
Resilience	Shadow Tolerance	Puccini Green Handbook (Gemeente Amsterdam, 2024b)	Feb 2024	Ratio shadow tolerance score to land area
	Drought Tolerance	Puccini Green Handbook (Gemeente Amsterdam, 2024b)	Feb 2024	Ratio drought tolerance score to land area
	Water Tolerance	Puccini Green Handbook (Gemeente Amsterdam, 2024b)	Feb 2024	Ratio water tolerance score to land area
Independent Variables				
Shannon's Index		Puccini Green Handbook (Gemeente Amsterdam, n.d.-a)	Aug 2025	-
Control Variables				
Population Density		(Gemeente Amsterdam, Onderzoek & Statistiek, n.d.)	2025	Number of residents per km^2 of land
Pollution Perception		WiA	2023	Average rating (1-10)
Socio-Economic Status		CBS	2023	Average rating (1-10)
Maintenance of Green Perception		WiA	2023	Average rating (1-10)

Table 1. Variables

The relationship between street tree diversity and both ecosystem services and ecological resilience was analyzed to assess whether areas with higher tree species diversity also provide greater ecosystem services. To determine which model is most adequate, BIC scores were calculated. The Log-Linear model had the lowest BIC score and was therefore ultimately chosen for the analysis.

This approach allowed for the evaluation of the influence of tree species diversity on ecosystem services and ecological resilience while accounting for potential confounding factors.

A series of tests were performed in Python to evaluate if assumptions of linear regression were met. Visual assessment of plots ensured the linearity and homoscedasticity of the data. Calculations of the Variance Inflation Factor (VIF) tests revealed the presence of multicollinearity between independent variables, resulting in less accurate coefficient values. To address this, the log-linear analysis was fitted with a ridge regression model from the python package Scikit-learn (Pedregosa et al., 2011). In such a case with multiple highly correlated variables, this helps prevent overfitting, resulting in a more stable and accurate model. Additional tests did not indicate any significant regression assumption violations, confirming that the data is appropriate for linear regression analysis.

3.2.2 Spatial analysis

A spatial analysis was conducted; Python was used to process and visualize the spatial relationship between urban trees and urban underground infrastructure in a spatial map. The underground infrastructure was chosen, since there is a lack of attention mentioned by Miron & Millward (2024) Jim (2003). Different methods were reviewed, the Puccini Methode (Gemeente Amsterdam, 2024b) and Boommonitor (Norminstituut Bomen, 2022), to determine the required space for a specific tree's roots to healthy tree growth. In combination with the public data on underground infrastructure like sewage systems and electric cables, a spatial analysis aims at identifying potential conflict areas within the inner city of Amsterdam, where spatial limitations constrain tree roots development.

The spatial model plotted every tree from the tree database in the inner city of Amsterdam. The 2D spatial location of the underground utility infrastructure was added to the spatial map. The utility infrastructure is divided into the sewage system and the low and middle voltage cables. These spatial components within the inner city of Amsterdam form the bases for the analysis.

To analyze the potential conflicts, a python script was run on the spatial map. In this script, every tree was given a radius with a determined boundary. The boundaries used were from the Puccini method and the Boommonitor. After creating the specific radius for every tree, the model finds if the tree has overlapped a utility line or multiple. If there is an overlap, the model determines that there is a risk of interference. The results will be shown in a spatial map and afterwards a final analysis is made to plot the total interferences per utility (sewage and power) in a graph.

4. Results

4.1 Data Results

The descriptive statistics of all variables are presented in Table 2. Figure 2 shows the biodiversity calculated through the Shannon´s Index per neighborhood in the center of Amsterdam.

Variable		**Mean**	**Median**	**Standard Deviation**	**Minimum**	**Maximum**
Dependent Variables						
Ecosystem Service	Cooling Effect	251.580	208.000	192.889	2.0	831.000
	Removing Pollutants	887.536	735.000	640.882	6.0	2486.000
	Supporting Biodiversity	167.855	140.000	123.162	1.0	472.000
Resilience	Shadow Tolerance	59.537	28.500	80.199	1.0	422.000
	Drought Tolerance	59.073	26.143	80.784	2.0	434.000
	Water Tolerance	59.003	18.000	68.969	1.0	334.000
Independent Variables						
Shannon's Index		2.101	2.118	0.588	0.0	3.283
Control Variables						
Population Density		14974.594	13444.000	6722.564	0.0	27999.000
Pollution Perception		5.435	5.405	0.712	3.6	7.000
Socio-Economic Status		6.938	7.000	0.532	5.2	7.900
Maintenance of Green		6.324	6.417	0.697	4.4	7.800

Table 2. Descriptive Statistics

Figure 2. Shannon´s Index per neighborhood in the center of Amsterdam.

The results from the regression analyses are represented in Table 3 and 4. The tables present the coefficients, denoted as β, standard errors as S.E., and p-values, for each predictor variable across the different dependent variables. Statistical significance is expressed with '*' symbols for p-values smaller than 0.05, smaller than 0.01 by '**', and smaller than 0.001 by '***'.

	Cooling Effect	R^2 = 0.586	R^2_{adj} = 0.547	Removing Pollutants	R^2 = 0.539	R^2_{adj} = 0.494	Supporting Biodiversity	R^2 = 0.525	R^2_{adj} = 0.479
	β	S.E.	p-value	β	S.E.	p-value	β	S.E.	p-value
Independent Variables									
Shannon's Index	0.508 ***	0.116	0.000014	0.468 ***	0.116	0.000056	0.482 ***	0.126	0.000118
Control Variables									
Resilience Score	0.227 ***	0.066	0.000623	0.137	0.053	0.009	0.131	0.057	0.016
Population Density	0.091	0.058	0.100	0.090	0.055	0.103	0.061	0.06	0.306
Pollution Perception	0.071	0.051	0.163	0.068	0.048	0.157	0.077	0.053	0.151
Socio-Economic Status	0.153	0.075	0.042	0.131	0.069	0.059	0.137	0.073	0.058
Maintenance of Green	0.003	0.054	0.956	0.001	0.048	0.978	-0.004	0.054	0.947

Table 3. Regression Results – Ecosystem Services

The R^2 and adjusted-R^2 values of these models (for cooling effect: $R^2 = 0.586$ and $R^2_{adj} = 0.047$, for removing pollutants: $R^2 = 0.539$ and $R^2_{adj} = 0.494$, for supporting biodiversity: $R^2 = 0.525$ and $R^2_{adj} = 0.479$) indicate that our analysis has meaningful explanatory power.

Across all three models, Shannon's Index was a highly significant positive predictor of ecosystem services. It shows strong effects on cooling effect ($\beta = 0.508$, $p < 0.001$), removal of pollutants ($\beta = 0.468$, $p < 0.001$), and supporting biodiversity ($\beta = 0.482$, $p < 0.001$). These results indicate that higher biodiversity is strongly associated with increased ecosystem service provisioning.

Among the control variables, the resilience score showed a positive and statistically significant association with the cooling effect ($\beta = 0.227$, $p < 0.001$), removal of pollutants ($\beta = 0.137$, $p < 0.001$), and supporting biodiversity ($\beta = 0.131$, $p < 0.001$). This suggests that more resilient urban tree systems may contribute to improved ecosystem services.

The other control variables of population density, pollution perception, and maintenance of green areas did not show statistically significant effects in any of the models. Interestingly, socio-economic status was found to be correlated with neighborhood cooling effect scores ($\beta = 0.137$, $p < 0.005$). This would indicate that trees found in wealthier neighborhoods are more likely to have a higher cooling effect. The fact that the control variables generally do not show significance suggests that the Shannon's Index and tree resilience explains most of the variation in the ecosystem service outcomes, and other variables do not add much in terms of explanatory power within this case. Biodiversity and resilience might be the primary drivers of the ecosystem services measured, while other social or perceptual factors play a smaller or limited role.

	Shadow Toler-ance	R^2 = 0.263	R^2_{adj} = 0.204	Drought Tolerance	R^2 = 0.355	R^2_{adj} = 0.304	Water Tol-erance	R^2 = 0.204	R^2_{adj} = 0.140
	β	S.E.	p-value	β	S.E.	p-value	β	S.E.	p-value
Independent Variables									
Shannon's Index	0.283 ***	0.103	0.006	0.361*	0.142	0.011	0.098	0.168	0.557
Control Variables									
Population Density	-0.057	0.074	0.436	-0.046	0.091	0.607	-0.209	0.121	0.083
Pollution Perception	0.066	0.055	0.229	0.034	0.085	0.69	-0.064	0.144	0.66
Socio-Eco-nomic Sta-tus	0.124	0.094	0.19	0.132	0.113	0.243	0.049	0.132	0.71
Mainte-nance of Green	0.192**	0.063	0.002	0.352 ***	0.095	0.000219	0.479 ***	0.138	0.000469

Table 4. Regression Results – Resilience Score

The models explain the different aspects of resilience to different degrees. The explanatory capabilities are strongest for the drought tolerance model ($R^2 = 0.355$, $R^2_{adj} = 0.304$), where around a third of the variation is accounted for. Shadow tolerance shows moderate explanatory power ($R^2 = 0.263$, $R^2_{adj} = 0.204$), and water tolerance is explained the least ($R^2 = 0.204$, $R^2_{adj} = 0.140$), meaning a substantial amount is influenced by factors that are not captured here.

The results show that Shannon's Index was a significant positive variable for shadow tolerance ($\beta = 0.283$, $p = 0.006$), indicating that areas with higher biodiversity tend to exhibit greater capacity to tolerate shaded conditions. A similar positive trend was observed for drought tolerance ($\beta = 0.361$, $p = 0.011$), indicating that areas with higher tree species diversity generally include species which are more resilient to drought conditions. No significant relationship was found between biodiversity and flooding tolerance ($\beta = 0.098$, $p = 0.557$).

As for the control variables, the perception of maintenance of green spaces showed a consistent and strong positive effect across all three resilience dimensions: shadow tolerance ($\beta = 0.192$, $p = 0.002$), drought tolerance ($\beta = 0.352$, $p < 0.001$), and water tolerance ($\beta = 0.479$, $p < 0.001$). This suggests that areas perceived to be more maintained usually have a higher overall resilience. This may highlight the importance of regular maintenance in creating a more urban resilient tree structure. Population density is shown to be negatively associated with water tolerance ($\beta = -0.209$), but this relationship is only marginally significant ($p = 0.083$). Pollution perception and socio-economic status did not show significant effects across any of the models.

4.2 Spatial Conflicts

The spatial analysis was performed to assess the conflict between urban trees currently and the underground utility network, specifically by quantifying the number of trees at risk across five different clearance scenarios. This analysis utilized the minimum required clear radius for each boundary (20, 40, 60, 80-year lifespan and the Puccini guideline) to determine the spatial interference. The interference shown in Table 5 directly translates to a clear restriction in the underground space required for the optimal root development, thereby limiting the trees' capacity to achieve their potential lifespan.

	Spatial boundary (m)	Trees with interference risk with underground infrastructure	% of total trees	Interferences Sewage	Interferences Power lines
Puccini Methode	2.0	1.615	14,78	814	888
20-year lifespan	1.5	1.129	10,33	577	585
40-year lifespan	1.7	1.307	11,96	666	694
60-year lifespan	2.1	1.714	15,69	881	939
80-year lifespan	2.6	2.275	20,82	1.183	1.339

Table 5. Results – Spatial Analysis

5. Discussion

5.1 Theoretical Discussion

The findings of this research align with the research conducted by van der Plas et al. (2016), as outlined in the theoretical framework. Van der Plas and colleagues argue that areas with a higher biodiversity exhibit a wider range of ecosystem services, in which species complement each other's weaknesses. This suggests that areas with higher biodiversity are a 'jack-of-all-trades' type of ecosystem where a group of diverse species serve ecosystem services on a moderate level but doesn't necessarily increase the excelling level of specific ecosystem service. The findings of this study reinforce this postulation as mentioned in the results section.

In contrast, the perspective proposed by Ridder (2007) does not align with the findings of this research. Ridder addresses that biodiversity is not inherently necessary for ecosystem services as the whole ecosystem services are not provided by species diversity, but by any group of species that meets certain basic functional criteria. From this viewpoint, ecosystem services depend more on functional traits of each species than on the overall species of richness.

Furthermore, patterns observed in urban tree diversity support the notion that higher species abundance contributes to the multiple ecosystem services. This finding corresponds with the argument of Isbell et al. (2022), who emphasized that a high level of species diversity is required to maintain multiple ecological functions at multiple times and places. Even though some species can be replaced by other species for a specific function under certain environmental conditions, a variety of species is needed to sustain various ecosystem services.

The theory emphasizes the significance of sufficient underground space needed for a healthy root system and tree, yet they recognize the influence of numerous other major factors (Watson, 2014). This limitation is strongly aligned by the present findings in the results. While the spatial requirements for a projected 60-year lifespan closely align with the Puccini guidelines, the actual average lifespan of an urban tree is closer to 40 years. This difference confirms that other factors beyond spatial clearance are critical constraints.

5.2 Limitations

Although this study aims to establish a theoretically strong foundation and a well-supported data analysis, there are certain limitations which need to be acknowledged. These limitations are discussed in this section.

First, the dataset used for mapping out the trees in the inner city of Amsterdam that was available online, only consists of public trees (with the same for underground infrastructure). As a result, trees located in private gardens, on company sites, and other private areas are not taken into account in this study, while these could take up a significant percentage of urban trees. The municipality of Amsterdam states that there are approximately 1 million trees, of which 300.000 are maintained by them (Gemeente Amsterdam, n.d.). Although this includes other neighborhoods in Amsterdam, it does emphasize that a large part of data is missed when relying solely on public data, which might result in biased outcomes. At the AMS Institute, a simultaneous study is being conducted using LiDAR technology to collect tree data throughout the entirety of Amsterdam, which would also include private trees (Bendixen, personal communication, October 2025). While that data is not available yet, this could be valuable for further research in the scope of this paper.

The quantifications for ecosystem services used in this research are simplified indices and do not fully capture real-world ecosystem functioning. Differences between the indicator and the real world might therefore be substantial. Calculations in this study are based solely on general information on the species of a tree, without regards to other factors such as tree age, canopy cover, location, proximity to water, seasonality, or other interactions with surroundings. These unaccounted variables likely influence the magnitude and variability of ecosystem services in reality. These variables were not included due to a lack of data availability but would have strengthened the analysis. Consequently, the results should be interpreted as a general indication rather than a precise estimation of ecosystem services.

The regression models conducted in this study reveal correlations between tree biodiversity and ecosystem services (or tree resilience), but don't provide any evidence of a causal relationship between the variables. There might be many other environmental or management factors influencing the ecosystem services, so the interpretations should be made carefully, without drawing conclusions on causal relationships.

The unit of analysis in this research is the neighborhood. This unit was chosen due to the nature of available data, but it does limit the research in certain keyways. Certain data used was collected at the individual level (individual inhabitant/tree) but subsequently aggregated and averaged to the neighborhood scale, which could introduce the risk of an ecological fallacy. This fallacy occurs when conclusions about individuals are drawn from group-level data. Due to the aggregation, no regard is given to the variance within the neighborhood, which could lead to incorrect conclusions. The results may not fully capture the diversity of characteristics of individuals within each neighborhood.

An additional limitation of having the neighborhood as the unit of analysis has to do with the neighborhood boundaries. For this research, neighborhood boundaries were aligned with the municipality's established boundaries. However, these administrative boundaries are to some extent arbitrary, especially with regards to ecosystem services and biodiversity scores. Ecological processes and environmental characteristics do not conform to administrative boundaries. Factors such as species distribution and abundance extend beyond neighborhood limits. The data therefore does not accurately capture the ecological dynamics that influence ecosystem service provision and resilience.

The data on ecosystem service scores and resilience scores is derived from the Puccini Green Handbook, in which each tree species receives an ordinal score of 1 to 3/4 for each service and each dimension of resilience. The handbook does not provide any explanation regarding how these scores were determined, nor does it clarify the precise meaning or quantitative difference between the scores. The rationale behind the scoring system remains unknown, and the interpretation of differences between species based on these scores is therefore inherently limited.

The spatial model concludes that there are potential interference risks for the required clear space a tree needs for a healthy growth for a specific lifespan but doesn't provide how significant the risk is. When conducting the Puccini boundaries, the model concludes many interferences with the guideline. However, in the guideline possible exemptions are stated to still protect the infrastructure within the clear space. The same can be said about the boundaries for specific lifespans. The theory implies that the clear space required is one of the many factors that determine the lifespan of an urban tree. So, the model cannot predict the expected lifespan of a tree even if the minimum clear space is provided. The model can only show if there is a potential risk of hinderances for tree growth for a specific lifespan.

Additionally, the model only conducts a 2D spatial analysis of the inner city of Amsterdam. This means that the depth of the infrastructure and tree roots were not considered. And the data used in the model is limited to only two infrastructures out of the many others in the underground.

6. Conclusion

This research highlights the interplay between urban tree diversity, resilience, and ecosystem services provisioning in the city center of Amsterdam. The findings support the 'jack-of-all-trades' idea by van der Plas et al. (2016), showing that areas with higher tree species diversity generally exhibit a higher degree of ecosystem services. This demonstrates the importance of creating a multifunctional urban forest capable of sustaining different environmental conditions as well as multiple services simultaneously. In this regard, the diversity of trees is critical. The current situation in the center of Amsterdam is problematic, as it substantially lacks the biodiversity of street trees. The area features a striking domination of one genus: the *Ulmus* (elm). Consequently, Santamour's (1990) diversity guidelines are significantly violated. In areas where Elm trees dominate, diversity is limited and can negatively affect the overall performance of the urban ecosystem. A larger range of species is preferred, not only to maintain ecological resilience through different conditions, but also to deliver a balanced and equally distributed set of services.

This research demonstrates that cities such as the city center of Amsterdam, which have a higher population density and that are protected by UNESCO heritage and green monumental status, pose specific challenges for street tree management. Maximizing resilience and service provisioning through diversity is important for effective urban forest management, but it also requires selecting species which have certain key functional traits and are suited for the dense urban landscape, as well as consideration of the local spatial aspects. The spatial analysis showed that the battle for space has multiple dimensions, and the underground dimension poses serious challenges to street tree management in the center of Amsterdam. Balancing these spatial considerations is crucial in promoting both resilience and service provision in a highly contested urban environment. Furthermore, there is a limitation for the scope of ecological interventions that aim to strengthen biodiversity and ecosystem service provision in cities that are protected as designated historic monuments.

The relationships identified in this research represent associations only, and before any policy recommendations can be made, the associative relationships should be further investigated to ensure the presence of causality. It is recommended that these associations are examined in greater detail. Future research should employ small scale approaches, measuring biodiversity, ecosystem services, and resilience on the micro-level. Not only would this greatly improve the explanatory power of the relationship but enhance sensitivity to case-specific characteristics.

7. References

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8. Statement on A.I. use

During the process of conducting this research and writing this paper, AI was used for several purposes. AI was primarily used for assisting in writing the code for running spatial analysis and making regression models. Additionally, AI was utilized in a couple of instances to find appropriate synonyms or to improve sentence coherence and structure. Furthermore, A.I. was applied for referencing in APA 7 style and organizing the sources in alphabetical order.

9. Appendix

9.1. Spatial Analysis

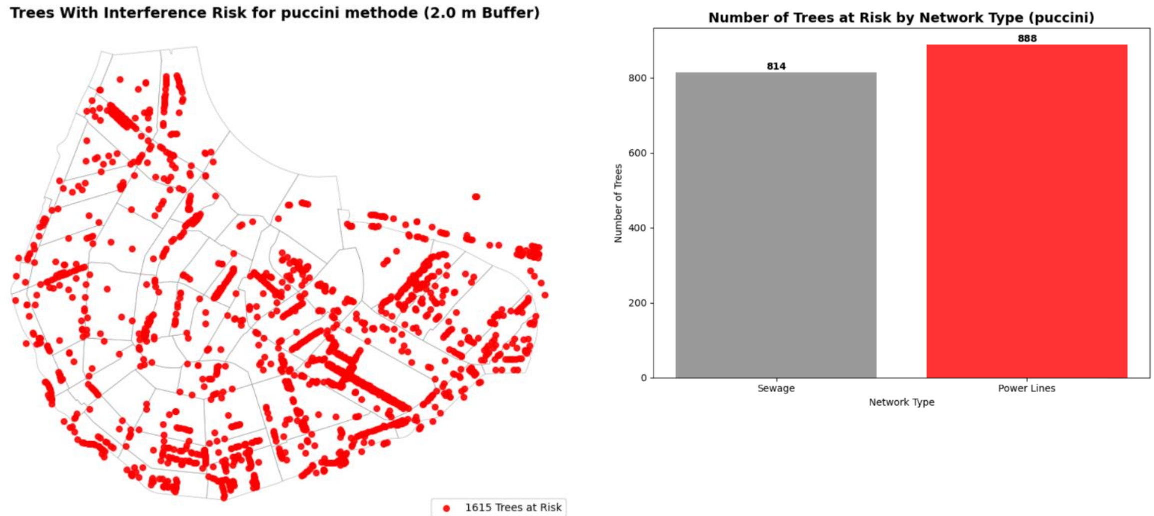

Figure 3. Distribution and Number of Trees at Risk by the Puccini Method

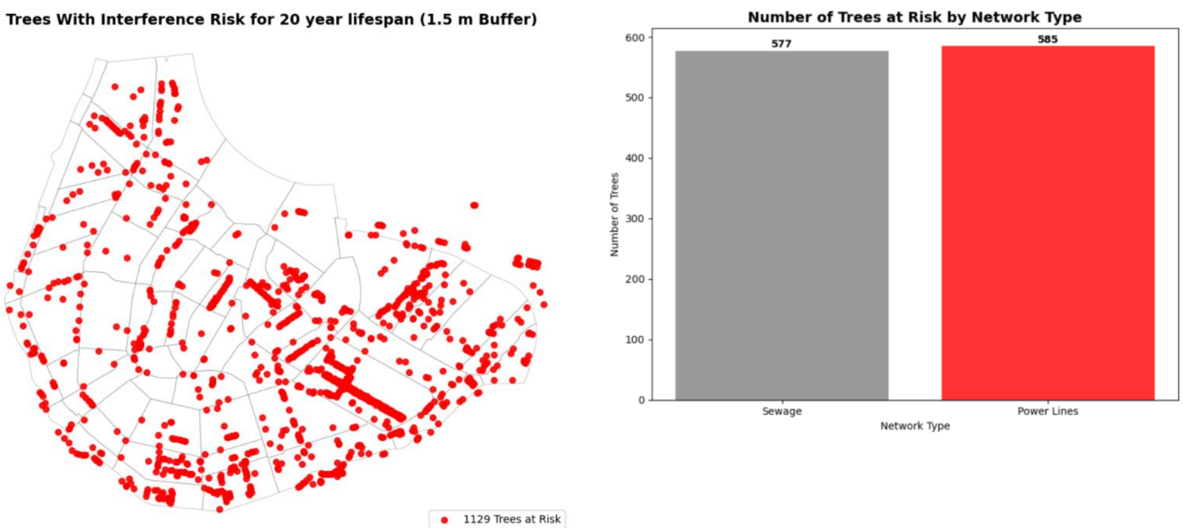

Figure 4. Distribution and Number of Trees at Risk of 20-year Lifespan

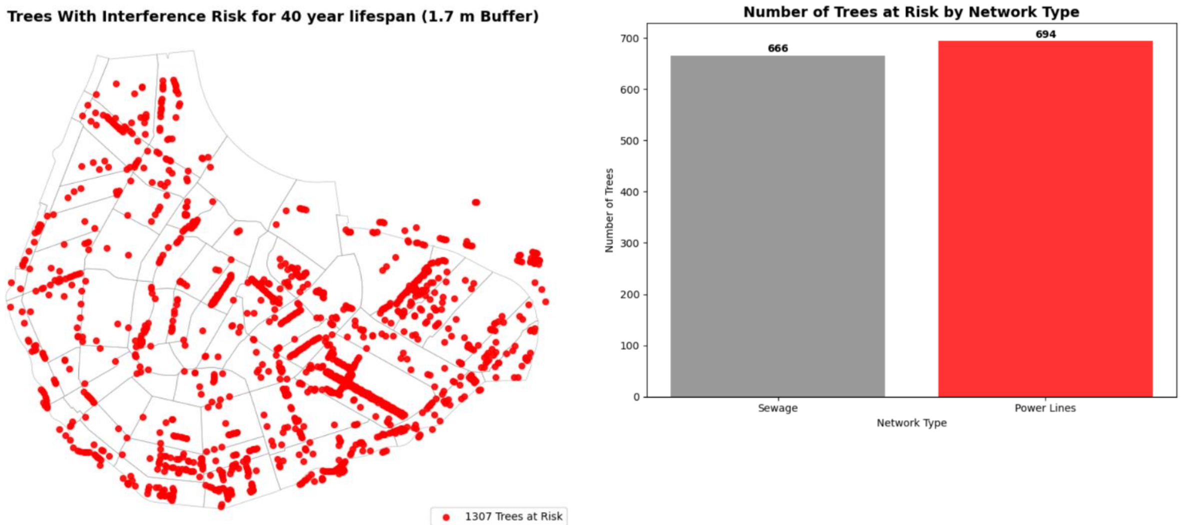

Figure 5. Distribution and Number of Trees at Risk of 40-year Lifespan

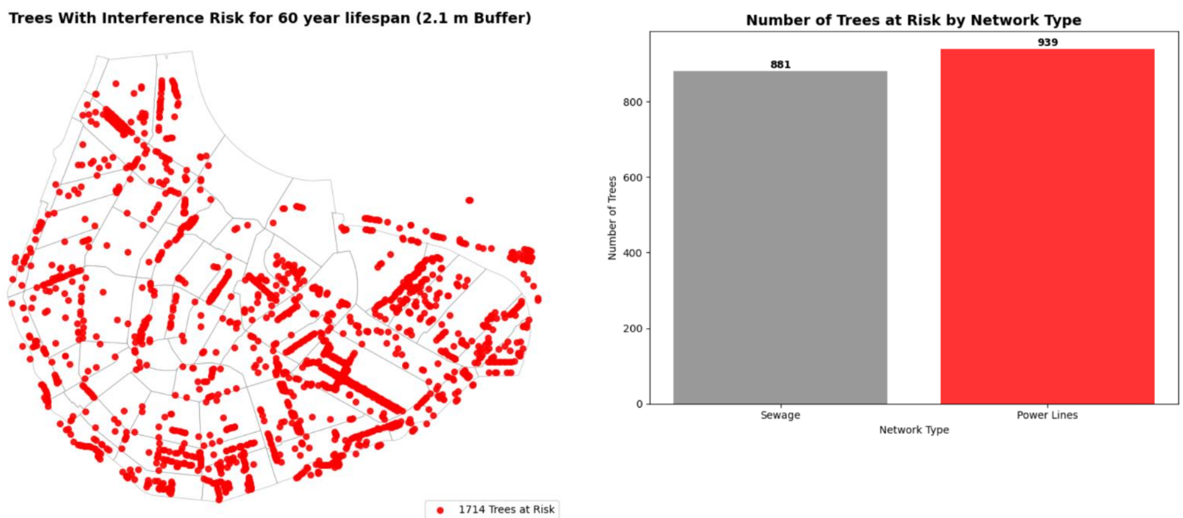

Figure 6. Distribution and Number of Trees at Risk of 60-year Lifespan

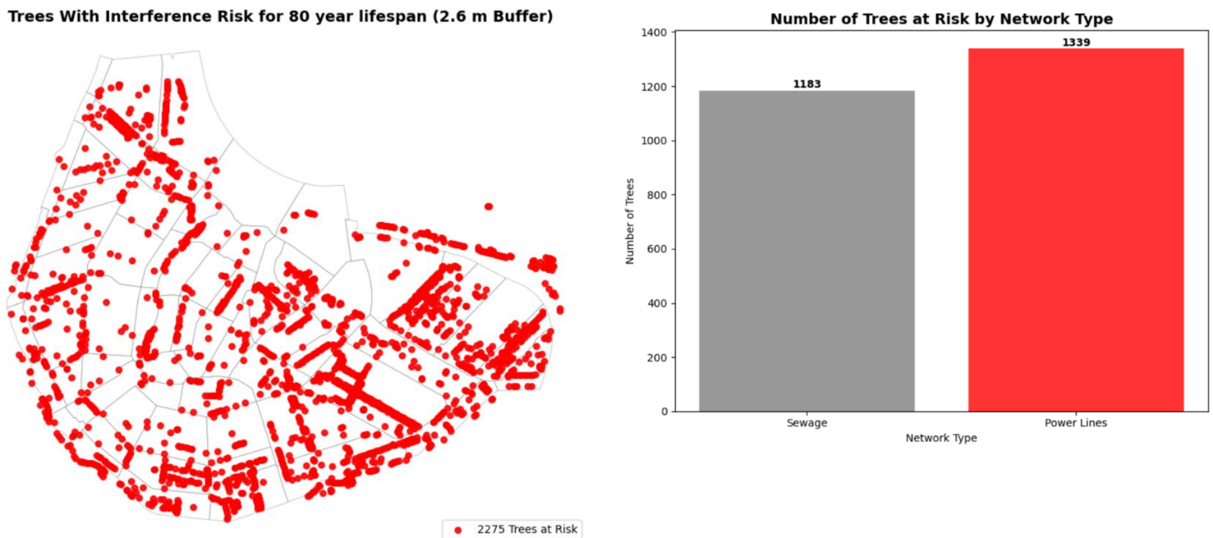

Figure 7. Distribution and Number of Trees at Risk of 80-year Lifespan